MULTI-STEP PROMPT WORKFLOW

Mini Project: Smart Attendance Management System

Name: Navya Reddy R

Registration Number: 99230040696

Scenario: Transforming a mini-project idea into a structured project proposal

1. Project Overview

Project Title: Smart Attendance Management System

Smart Attendance Management System is a simple web-based application that helps teachers record student attendance, maintain attendance records, and generate basic attendance reports. The project aims to reduce manual attendance work and provide a clear view of student attendance information.

Suggested Technology: HTML, CSS, JavaScript, Java, MySQL

Users: Students, teachers, and administrators

Core Features: Student entry, attendance marking, attendance search, daily attendance records, database storage, attendance history, and basic attendance reports.

2. Raw Project Notes

Many educational institutions maintain attendance details manually in registers or separate spreadsheets. It becomes difficult to track daily attendance, check individual student attendance history, and prepare attendance reports manually. The proposed Smart Attendance Management System should allow teachers to add student details, mark daily attendance, search attendance records, save information in a database, and view simple attendance reports. The interface should be easy for beginners to use.

STEP 1 — ANALYSIS & PROJECT PLANNING

Exact Prompt Used

> Act as an expert software project analyst and technical documentation writer.
>
> I want to develop a mini project called "Smart Attendance Management System".
>
> Analyze the raw project notes below and produce:
>
> 1. The main problem statement.
>
> 2. Exactly 4 key project objectives.
>
> 3. The target users.
>
> 4. Exactly 6 core functional requirements.
>
> 5. A simple technology stack using HTML, CSS, JavaScript, Java, and MySQL.
>
> 6. A structured outline for a 700-word mini-project proposal.
>
> Do not invent advanced features that are not supported by the notes.

Output format:

- Problem Statement

- Objectives

- Target Users

- Functional Requirements

- Technology Stack

- Proposal Outline

Raw notes:

Many educational institutions maintain attendance details manually in registers or separate spreadsheets. It becomes difficult to track daily attendance, check individual student attendance history, and prepare attendance reports manually. The proposed Smart Attendance Management System should allow teachers to add student details, mark daily attendance, search attendance records, save information in a database, and view simple attendance reports. The interface should be easy for beginners to use.

Intermediate AI Response — Live Execution Record

Paste the actual Step 1 response generated by the assigned AI interface here.

STEP 2 — PROPOSAL GENERATION

Exact Prompt Used

Excellent. Using only the project analysis, objectives, requirements, technology stack, and proposal outline from your previous response, write a complete mini-project proposal for the Smart Attendance Management System.

Strict constraints:

1. Maximum 700 words.

2. Use a professional academic tone.

3. Include: Introduction, Problem Statement, Objectives, Proposed Solution, Key Features, Technology Stack, Expected Benefits, and Conclusion.

4. Explain how the system works in simple terms.

5. Do not introduce features or technologies that were not included in Step 1.

6. Use short paragraphs of no more than 3 sentences each.

7. Include this simple text-based workflow:

User → Web Interface → Java Application → MySQL Database → Attendance Report.

8. Make the proposal suitable for a college mini-project submission.

Output only the first draft of the proposal.

Intermediate AI Response — Live Execution Record

Paste the actual Step 2 response generated by the assigned AI interface here.

STEP 3 — CRITIQUE & REFINEMENT

Exact Prompt Used

Act as a strict senior project reviewer. Critically evaluate the proposal you just wrote against the following checklist:

1. Is it 700 words or fewer?

2. Does it contain all required sections?

3. Are exactly 4 objectives represented?

4. Are the six core functional requirements from Step 1 accurately reflected?

5. Does it use only the approved technology stack: HTML, CSS, JavaScript, Java, and MySQL?

6. Is the workflow clear and technically consistent?

7. Are unsupported or exaggerated claims avoided?

8. Is the language suitable for an academic mini-project submission?

9. Are paragraphs limited to a maximum of 3 sentences?

10. Is the conclusion clear?

First, provide a "Self-Critique" with a bullet-by-bullet evaluation of all 10 criteria. Identify any weaknesses and explain how they should be fixed.

Then provide "Final Polished Mini-Project Proposal". Correct every issue you identify while preserving the useful content from the previous draft.

The final proposal must remain within 700 words and satisfy all 10 criteria. Do not skip the self-critique.

Intermediate AI Response — Live Execution Record

Paste the actual Step 3 self-critique and final proposal generated by the assigned AI interface here.

4. Chaining Dependency

Step 1 → Step 2 → Step 3

Step 1 transforms raw project notes into a structured analysis. Step 2 explicitly uses the output from Step 1 to generate a complete project proposal. Step 3 explicitly evaluates the Step 2 proposal against a defined checklist and transforms it into a corrected final version.

5. Critique Gate

Step 3 is the required critique gate. The AI must first review its own previous proposal, identify weaknesses against ten specific criteria, and then produce a refined final version that addresses the identified issues.

6. Live Execution Evidence

Run all three prompts sequentially in the same assigned AI interface. Add screenshots showing Prompt 1 and its response, Prompt 2 and its response, and Prompt 3 with the self-critique and final response. If the platform provides a shareable chat link, include it in the submission.

7. Assessment Checklist

Requirement — Evidence in this workflow

3+ chained prompts — Three sequential prompts are provided.

Explicit dependency — Steps 2 and 3 explicitly use the preceding outputs.

Critique gate — Step 3 contains a 10-point review and correction stage.

Mini-project scenario — Smart Attendance Management System is used as the project.

Live testing — Actual outputs and screenshots must be inserted after running the prompts.

8. Submission Note

Replace each 'Live Execution Record' placeholder with the exact AI-generated output obtained by running the three prompts in sequence. Add screenshots or the chat link as required by the assessment instructions.